System Analysis and Modelling Document

Augmented Reality Music Production & Digital Audio Engineering Software Workstation System

(built on a virtual 3D shaped music studio)

CMPU 2020 Software Engineering 2 – C24329646

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System Overview

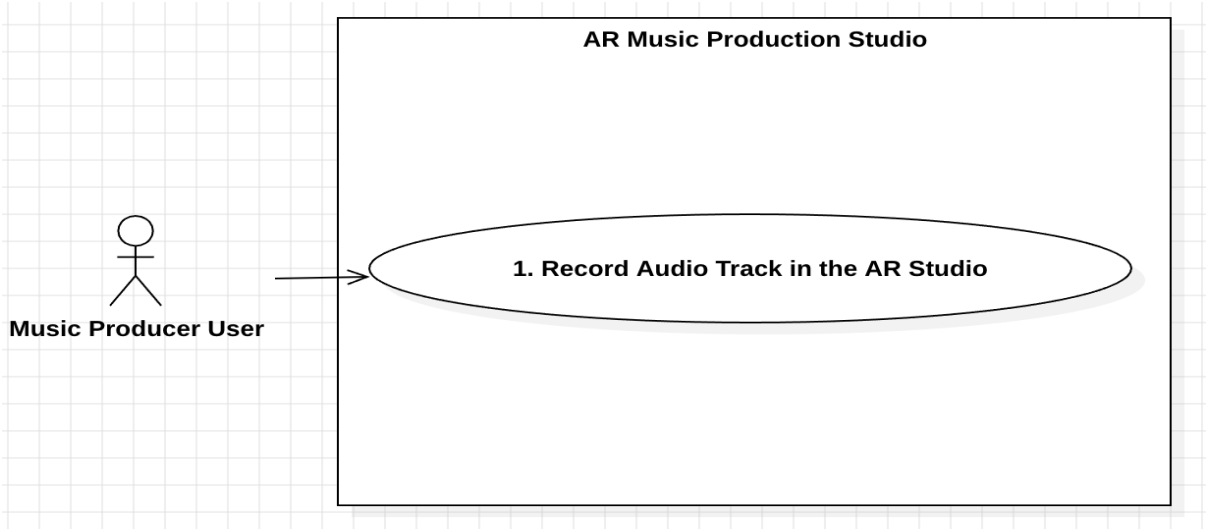
This Software Engineering 2 project will provide a potential Powered Music Production Tool, such as DAW Plugin or Standalone App, in a better, more efficient, professional, and accessible way, based on an augmented reality version of a 3D shaped studio room, for professionals and beginners to be able to create, hold, and secure multiple tracks at the same time, with the greatest audio recording quality, most advanced compressors, effects, and plugins, with a methodical editing, structure, and stability, as well as providing interactivity and social aspects to enhance their experience.

Thinking of a 3D shaped studio room, not a screen:

- height: each track is a floating row such as a physical object in space
- length/width: mixer act as real console/control panel unit with each channel rendered/generated as vertical faders/volume interface that you can reach out/grab with pinch gestures
- depth: spatial audio, maybe with instruments around you in physical space.

11. USE CASE ITERATION 1 - AR Music Production Studio

Description: The following StarUML Use Case Iteration 1 is focused on "Record Audio Track in the AR Studio", containing one actor, the music producer user. This overall use case will be broken down into parts in iteration 2.



USE CASE	1	Use Case - Iteration 1 Record Audio Track in the AR Studio
Description of Goal in Context	A music producer uses their smartphone to record, develop, and produce an audio track in a virtual 3D studio using a mixer anywhere at any time.	
Preconditions	• The system app needs to be installed; • The smartphone needs to contain a camera and microphone permissions enabled; • The system app needs to have a project opened for creation.	
Post Conditions, Success End Condition	• The recorded and finished track audio needs to appear as a floating row in the 3D studio, its mixer settings applied, and the project completely saved.	
DESCRIPTION of Scenario	• Music producer points their smartphone at the room; • System displays the 3D studio around them; • Music producer creates a new track and records a vocal or instrument take; • Track appears as a floating row, and channel fader appears on the 3D console;	

	• Music producer pinches and drags the fader to set the volume, adjusts the pan, and plays the track back to check the mix; • System completely saves the project.

Main Flow

Step	Action	Alternate
1.1	User opens the app	
1.2	User authenticates their registration	
1.3	User creates a new project	
1.4	System activates the camera, requesting permission	EF 1.4
1.5	User points camera at the room	
1.6	System displays 3D studio around user in augmented reality version	
1.7	User adds a new audio track	
1.8	System shows audio track as a floating row, creating a mixer channel on the 3D console	
1.9	User presses record to document a vocal or instrumental take	
1.10	System records the audio documented	EF 1.10
1.11	System displays the waveform on the track	
1.12	User pinches the channel fader with a gesture, adjusting volume and pan	AVF 1.12
1.13	System applies audio changes in real time	
1.14	User presses play to review the mix	
1.15	System completely automatically autosaves the project	

EXCEPTIONS or ERROR Flow

Description: Camera and microphone permissions denied or unavailable

Step	Branching Action	Alternate
1.4.1	System informs the user with an error message that the camera or microphone permissions are denied or unavailable	

1.4.2	System prompts the user to open settings to try again or to connect an external audio interface	
EXCEPTIONS or ERROR Flow **Description:** Storage full or app crashed during recording		
Step	**Branching Action**	**Alternate**
1.10.1	System informs the user with an error message that the storage is full, or the app crashed during recording	
1.10.2	System automatically saves and stores the recording track until almost a minute before the lost	
1.10.3	System prompts the user to try again, after they cleaned up storage	
ALTERNATIVE or VARIATION Flow **Description:** User prefers touch or mouse controls		
Step	**Branching Action**	**Alternate**
1.12.1	In terms of accessibility, user prefers to use physical controls, instead of pinch and dragging gestures	
1.12.2	User types or speaks their choice to the system	
1.12.3	System listens and switches controls	

Non-functional Requirements for each Use Case can be added in the Table below. Only consider 2 or 3 non functional requirements from the list in the table.

Non-functional requirements, management issues and decisions required to be made, can be identified in the following table.

From the table below, **choose <u>a limited number</u> of appropriate non-functional requirements**relevant to the Use Case.

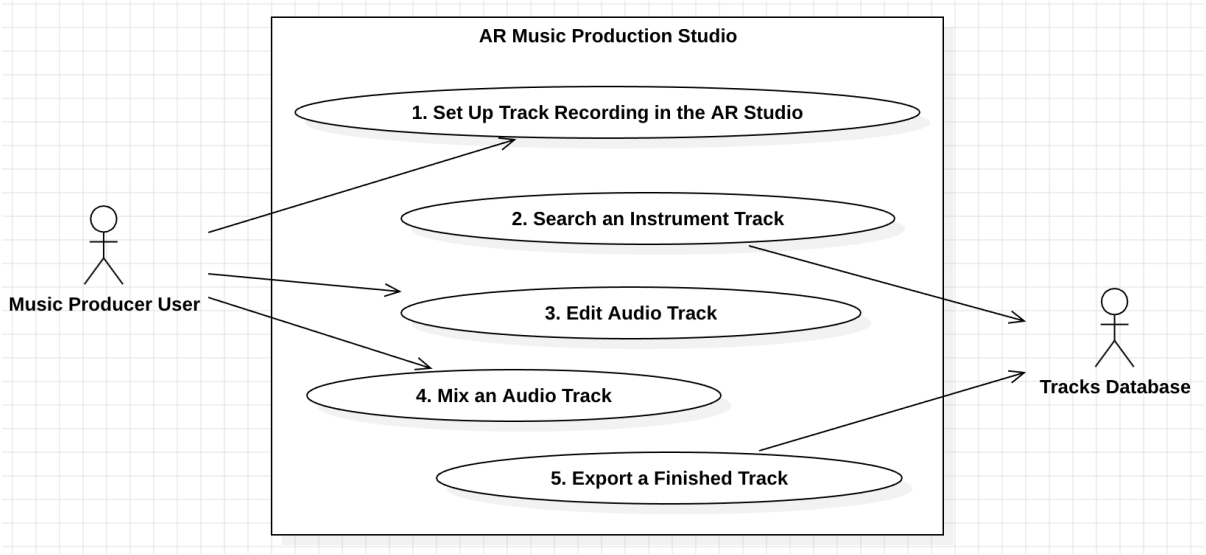
Non Functional Requirements can be categorised as

- Product related
- Organisation related, process and approaches set by the company
- External, imposed by outside bodies

RELATED INFORMATION	Use Case: 1	Use Case - Iteration 1 Record Audio Track in the AR Studio
Priority:	High since it is core feature	
Product: Reliability	Autosaving tool means a crash loses no more than almost one minute of unsaved work.	
External: Legislation	User recordings are strictly personal, including all their personal data, and must follow GDPR.	
Frequency	System used multiple times per session, in almost every session.	
Channels to actors	Interactive channels are the smartphone touch screen, camera, and microphone.	

12. USE CASE 1 - AR Music Production Studio (5 Use Cases)

Description: In the following StarUML use case, the system boundary is called AR Music Production Studio, the human actor is represented by the music producer user, and there are two external databases, one for music producer, and one for tracks.



1. SET UP TRACK RECORDING IN THE AR STUDIO

USE CASE	1	Use Case 1 Set Up Track Recording in the AR Studio
Description of Goal in Context	A music producer uses their smartphone to record and add an audio track in a virtual 3D studio, anywhere at any time.	
Preconditions	• The system app needs to be installed; • The smartphone needs to contain a camera and microphone permissions enabled; • The system app needs to have a project opened for creation.	
Post Conditions, Success End Condition	• The recorded track audio needs to appear as a floating row in the 3D studio, and the project completely saved.	
DESCRIPTION of Scenario	• Music producer points their smartphone at the room; • System displays the 3D studio around them; • Music producer creates a new track and records a vocal or instrument take; • Track appears as a floating row, and channel fader appears on the 3D console; • System completely saves the project.	

Main Flow		
Step	**Action**	**Alternate**
1.1	User opens the app	
1.2	User authenticates their registration	
1.3	User creates a new project	
1.4	System activates the camera, requesting permission	EF 1.4
1.5	User points camera at the room	
1.6	System displays 3D studio around user in augmented reality version	
1.7	User adds a new audio track	
1.8	System shows audio track as a floating row, creating a mixer channel on the 3D console	
1.9	User presses record to document a vocal or instrumental take	AVF 1.9
1.10	System records the audio documented	EF 1.10
1.11	System displays the waveform on the track	
1.12	User presses play to review the mix	
1.13	System completely autosaves the project	
EXCEPTIONS or ERROR Flow **(do I say "Exception at Step 4 of Use Case 1:"?)** **Description:** Camera and microphone permissions denied or unavailable		
Step	**Branching Action**	**Alternate**
1.4.1	System informs the user with an error message that the camera or microphone permissions are denied or unavailable	
1.4.2	System prompts the user to open settings to try again or to connect an external audio interface	

EXCEPTIONS or ERROR Flow **Description:** Storage full or app crashed during recording		
1.10.1	System informs the user with an error message that the storage is full, or the app crashed during recording	
1.10.2	System automatically saves and stores the recording track until almost a minute before the lost	
1.10.2	System prompts the user to try again, after they cleaned up storage	
ALTERNATIVE or VARIATION Flow **Description:** User chooses an external audio interface		
Step	**Branching Action**	**Alternate**
1.9.1	User chooses an external audio interface, instead of the phone microphone	
1.9.2	System routes the recording from that device	

2. SEARCH AN INSTRUMENT TRACK

USE CASE	2	Use Case 2 Search an Instrument Track
Description of Goal in Context	A music producer searches for an instrument track, such as a piano roll, bass line, or synthesizer, by name, instrument type, or genre, previewing it, and adding it to the open project in the AR Studio.	
Preconditions	• The system app needs to be installed; • An internet connection is available; • The system app needs to have a project opened for creation.	
Post Conditions, Success End Condition	• The selected instrument track appears as a floating row in the 3D studio, with a matching channel fader on the 3D console, and the project completely saved.	
DESCRIPTION of Scenario	• Music producer opens the instrument track database panel inside the AR studio; • Music producer types or speaks a search item, with options to filter instrument, or genre; • System queries Tracks Database; • System displays the matching tracks as floating cards in the 3D space; • Music producer previews a track through the spatial audio; • Music producer chooses the track to add it to the project; • System adds the track as a floating row, creating a mixer channel.	

Main Flow

Step	Action	Alternate
2.1	User opens the app	
2.2	User opens the instrument track database panel in the AR studio	
2.3	User searches an item, selecting optional filters	
2.4	System sends the search request to the Track Database	EF 2.4
2.5	Track Database returns the matching tracks	EF 2.5
2.6	System displays the results as floating cards around the user in the 3D studio	

2.7	User selects a card which system previews it through spatial audio	AVF 2.7
2.8	User chooses a track to add it to the project	
2.9	System adds the track as a floating row, creating a channel fader on the 3D console	
2.10	System completely autosaves the project	

EXCEPTIONS or ERROR Flow

Description: Tracks database unreachable or no internet connection

Step	Branching Action	Alternate
2.4.1	System informs the user with an error message that the database is unreachable or there is no internet connection	
2.4.2	System offers to open locally stored projects in offline mode	
2.4.3	System automatically retries the connection	

EXCEPTIONS or ERROR Flow

Description: No matching tracks found

Step	Branching Action	Alternate
2.5.1	System informs the user with an error message that there are no matching tracks found	
2.5.2	System offers to do a broader search term, removing filters	

ALTERNATIVE or VARIATION Flow

Description: User prefers touch or mouse controls

Step	Branching Action	Alternate
2.7.1	In terms of accessibility, user prefers to use physical controls, instead of pinch and dragging gestures	
2.7.2	User types or speaks their choice to the system	

2.7.3	System listens and switches controls	

3. EDIT AUDIO TRACK

USE CASE	3	Use Case 3 Edit Audio Track
Description of Goal in Context	A music producer edits recorded audio clips, such as cut, copy, move, trim, loop, undo, and redo, on a track in the 3D Studio, anywhere at any time.	
Preconditions	• The system app needs to be installed; • The user opened a project, and the AR Studio is displayed; • The user already recorded audio of at least one track.	
Post Conditions, Success End Condition	• The edit is applied and the waveform is updated; • The user project is autosaved with the edited clip into the Music Producer Database.	
DESCRIPTION of Scenario	• Music producer selects a floating track row, choosing a clip area applying an edit; • System updates the waveform immediately; • Music producer previews the result; • System automatically saves the project; • Producer has the option to undo it if not wanted.	

Main Flow

Step	**Action**	**Alternate**
3.1	User selects an audio track, represented as floating row	
3.2	System highlights the track, showing it in waveform	
3.3	User selects a clip area	EF 3.3
3.4	User chooses an edit action	EF 3.4
3.5	System applies the edit action, updating the waveform	
3.6	User presses play to preview the edit	
3.7	System plays the edited track	

3.8	User accepts the edit, or presses undo and redo	AVF 3.8
3.9	System automatically saves the project	
EXCEPTIONS or ERROR Flow **Description:** No clip area selected		
Step	**Branching Action**	**Alternate**
3.3.1	System informs the user with an error message that there is no clip area selected	
3.3.2	System prompts the user to select a clip before choosing an edit action	
EXCEPTIONS or ERROR Flow **Description:** Edit is invalid or exceeds the clip bounds		
Step	**Branching Action**	**Alternate**
3.4.1	System informs the user with an error message that there the edit selected is invalid or exceeds the clip bounds	
3.4.2	System keeps the original clip, and prompts the user to select another one	
ALTERNATIVE or VARIATION Flow **Description:** User presses undo or redo		
Step	**Branching Action**	**Alternate**
3.8.1	User presses undo or redo option	
3.8.2	System reverts or reapplies the last edit action, refreshing the waveform	

4. MIX AUDIO TRACK

USE CASE	4	Use Case 4 Mix Audio Track
Description of Goal in Context	A music producer sets volume, pan, and effects for each track using the 3D mixer console, placing spatial sound in 3D space, anywhere at any time.	
Preconditions	• The system app needs to be installed; • The user opened a project, and the AR Studio is displayed; • The user already recorded audio of at least one track.	
Post Conditions, Success End Condition	• The volume, pan, and effect settings are applied to the track; • The mixer settings are automatically saved in the project.	
DESCRIPTION of Scenario	• Music producer opens the 3D console where each track has a channel with a vertical fader; • Music producer pinches and drags fader to set the level, adjusting pan, and effects such as EQ, compression, and reverb, playing back with spatial audio; • System applies each change in real time and saves the settings.	

Main Flow

Step	**Action**	**Alternate**
4.1	User opens the 3D mixer console	
4.2	System displays a channel fader for each track	EF 4.2
4.3	User pinches and drags a fader to set the volume	EF 4.3, AVF 4.3
4.4	System applies the volume change in real time	
4.5	User adjusts the pan, positioning the track in 3D space	
4.6	User adds an effect to the channel	
4.7	System applies the effect, displaying its settings	

4.8	User presses play to check the mix	
4.9	System plays the mix with the spatial audio positions applied	
4.10	System automatically saves the project	
EXCEPTIONS or ERROR Flow **Description:** AR tracking lost		
Step	**Branching Action**	**Alternate**
4.2.1	System informs the user with an error message that AR tracking is lost	
4.2.2	System prompts the user to reconnect	
4.2.3	System switches to the 2D studio view, keeping all mixer settings unchanged	
EXCEPTIONS or ERROR Flow **Description:** Gesture not recognised		
Step	**Branching Action**	**Alternate**
4.3.1	System informs the user with an error message that AR tracking is lost	
4.3.2	System prompts the user to retry or use touch controls	
ALTERNATIVE or VARIATION Flow **Description:** User prefers touch or mouse controls		
Step	**Branching Action**	**Alternate**
4.3.3	In terms of accessibility, user prefers to use physical controls, instead of pinch and dragging gestures	
4.3.4	User types or speaks their choice to the system	
4.3.5	System listens and switches controls	

5. EXPORT A FINISHED TRACK

USE CASE	5	Use Case 5 Export a Finished Track
Description of Goal in Context	A music producer exports the finished mix as an audio file, storing a copy in the Tracks Database, so it can be accessed, shared, or published anywhere at any time.	
Preconditions	• The system app needs to be installed; • The user opened a project, and the AR Studio is displayed; • The user already recorded and mixed audio of at least one track. • The user has an internet connection available and stable to store the copy.	
Post Conditions, Success End Condition	• The audio file is created in the chosen format, being saved on the device; • The audio file copy is stored in the Tracks Database, creating a link available to the user.	
DESCRIPTION of Scenario	• Music producer selects export option, choosing a format such as WAV, MP3, or AIFF, starting the action; • System renders the mix to an audio file, sending a copy to the Tracks Database; • System stores the file, confirming by a message displaying option to share.	

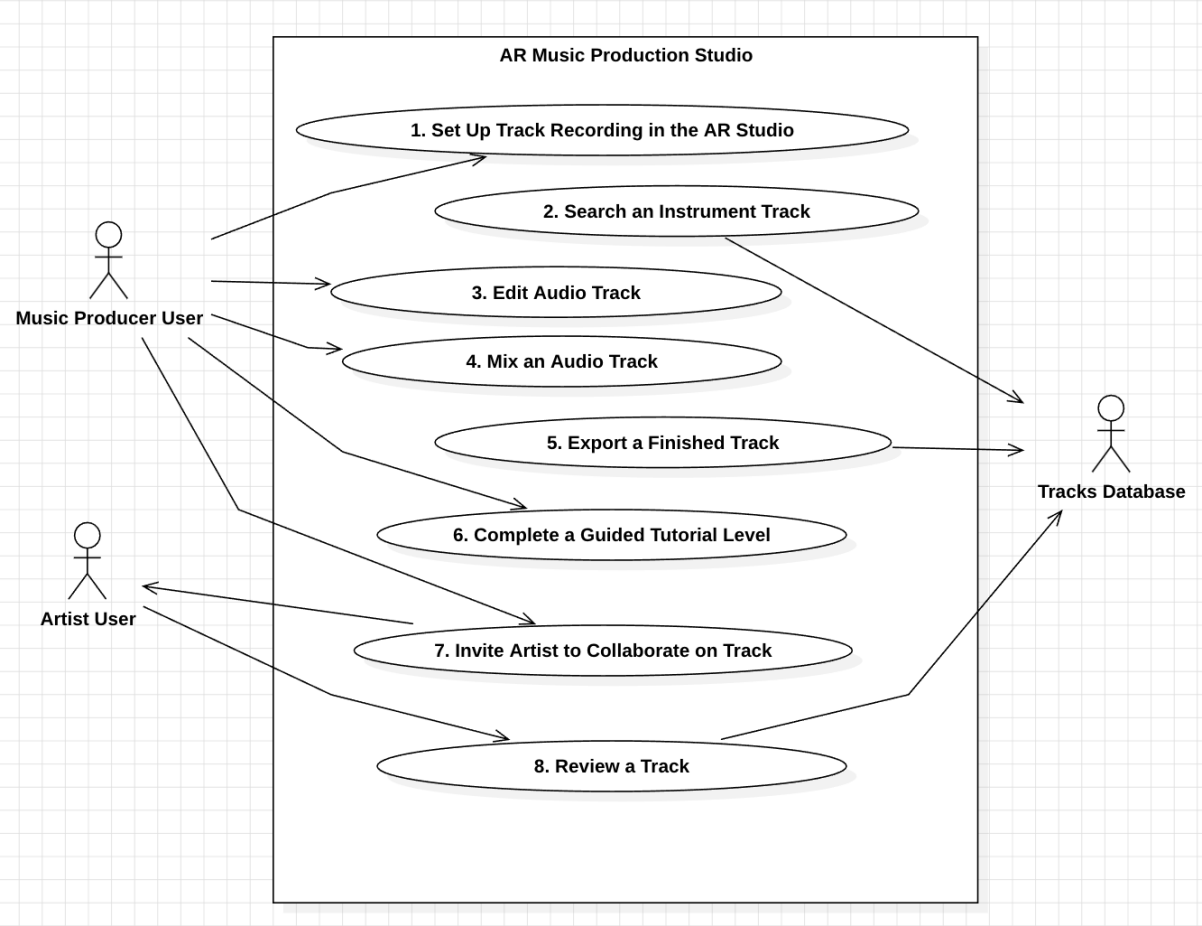
Main Flow

Step	Action	Alternate
5.1	User selects export option	
5.2	System displays the export option, such as format, quality, full mix, or specific systems	
5.3	User chooses a format, pressing export option	AVF 5.3
5.4	System renders the mix to an audio file, showing progress bar	EF 5.4
5.5	System sends a copy of the file to the Tracks Database	EF 5.5

5.6	Tracks Database stores the file and confirms with a record ID	
5.7	System saves the file on the device, displaying a confirmation option to share	
EXCEPTIONS or ERROR Flow **Description:** Export fails midway		
Step	**Branching Action**	**Alternate**
5.4.1	System informs the user with an error message that the export fails midway	
5.4.2	System deletes the partial file	
5.4.3	System prompts the user to try again	
EXCEPTIONS or ERROR Flow **Description:** Tracks Database unreachable		
Step	**Branching Action**	**Alternate**
5.5.1	System informs the user with an error message that the tracks database is unreachable	
5.5.2	System keeps the file on the device, queueing the upload	
5.5.3	System prompts the user to try again, checking if the connection is available	
ALTERNATIVE or VARIATION Flow **Description:** User chooses to export a single part of the track		
Step	**Branching Action**	**Alternate**
5.3.1	User chooses to export only a single part of the track, instead of the full mix	
5.3.2	System renders only the selected track	

12. ITERATION 3 (how to properly name between iteration, use case, narrative)

Description: In the following StarUML use case, the system contains now two users, and one database. **(what exactly in this description?)**



6. COMPLETE A GUIDED TUTORIAL LEVEL

USE CASE	6	Use Case 6 Complete a Guided Tutorial Level
Description of Goal in Context	A music producer follows a guided tutorial level in the virtual 3D studio, learning a production skill through gameplay, with target goals and achievements, anywhere at any time.	
Preconditions	• The system app needs to be installed; • The user opened a project, and the AR Studio is displayed.	
Post Conditions, Success End Condition	• The tutorial level is marked as completed; • The user progress and achievements are updated and saved.	

DESCRIPTION of Scenario	• Music producer opens the tutorial menu inside the AR studio and chooses a level; • System shows the goal of the level, guiding user through each task in the 3D studio; • Music producer performs each task, such as recording, editing, or mixing a practice track; • System checks the result, giving feedback; • System marks the level as completed, awarding an achievement, and saving the progress.

Main Flow

Step	Action	Alternate
6.1	User opens the tutorial menu in the AR studio	
6.2	User chooses a tutorial level	
6.3	System displays the level goal and the first task as a guide in the virtual 3D studio	
6.4	User performs the task, such as adding a track, adjusting a fader, or applying an effect	EF 6.4, AVF 6.4
6.5	System checks the result of the task	EF 6.5
6.6	System shows positive feedback, displaying the next task	
6.7	User repeats steps until all tasks of the level are done, displaying a complete message	
6.8	System marks the level as completed, awarding an achievement	
6.9	System automatically saves the progress on user profile	

EXCEPTIONS or ERROR Flow

Description: Task not completed correctly

Step	Branching Action	Alternate
6.5.1	System informs the user with an error message that the task was not completed correctly	

6.5.2	System displays a hint on how to complete the specific task	
6.5.3	System prompts the user to try again	
EXCEPTIONS or ERROR Flow **Description:** AR tracking lost during the tutorial		
Step	**Branching Action**	**Alternate**
6.4.1	System informs the user with an error message that AR tracking is lost	
6.4.2	System switches to the 2D studio view, keeping the tutorial progress unchanged	
6.4.3	System prompts the user to reconnect to continue the level	
ALTERNATIVE or VARIATION Flow **Description:** User chooses to leave the tutorial level and continues later		
Step	**Branching Action**	**Alternate**
6.4.4	User chooses to exit the tutorial level before finishing the level	
6.4.5	System automatically saves the progress up to the last completed task	
6.4.6	System prompts the user to resume the level from that task next time	

7. INVITE ARTIST TO COLLABORATE ON TRACK

USE CASE	7	Use Case 7 Invite Artist to Collaborate on Track
Description of Goal in Context	A music producer invites an artist to collaborate on a track of an open project in the virtual AR Studio, so the artist can access and review it, anywhere at any time.	
Preconditions	• The system app needs to be installed; • The user opened a project, and the AR Studio is displayed; • The user already recorded audio of at least one track; • An internet connection is available.	
Post Conditions, Success End Condition	• The artist receives and accepts the invitation; • The artist has access to the shared track, with the chosen permission; • The project is completely saved.	
DESCRIPTION of Scenario	• Music producer opens the collaboration panel inside the virtual AR studio, selecting a track; • Music producer enters the artist's name or email, choosing a permission, such as review only or review and comment; • System sends the invitation to the artist; • Artist receives and accepts the invitation; • System gives the artist access to the track, notifying the producer; • System automatically saves the project.	

Main Flow

Step	Action	Alternate
7.1	Music producer user opens the collaboration panel in the AR studio	
7.2	Music producer user selects the track to share	
7.3	Music producer user enters the artist's name or email, choosing a permission	
7.4	Music producer user presses send invitation	EF 7.4
7.5	System sends invitation to the artist	EF 7.5

7.6	Artist receives invitation, accepting it	AVF 7.6
7.7	System gives the artist access to the track, with the chosen permission	
7.8	System notifies the music producer user that the invitation was accepted	
7.9	System automatically saves the project	
EXCEPTIONS or ERROR Flow **Description:** Artist not found or invalid contact		
Step	**Branching Action**	**Alternate**
7.4.1	System informs the user with an error message that the artist was not found or the contact is invalid	
7.4.2	System prompts the user to re-check the name or email, allowing to try again	
EXCEPTIONS or ERROR Flow **Description:** No internet connection		
Step	**Branching Action**	**Alternate**
7.5.1	System informs the user with an error message that there is no internet connection	
7.5.2	System keeps the invitation queued	
7.5.3	System automatically retries sending when the connection is available	
ALTERNATIVE or VARIATION Flow **Description:** Artist declines the invitation		
Step	**Branching Action**	**Alternate**
7.6.1	Artist declines the invitation	
7.6.2	System notifies the user that the invitation was declined	

7.6.3	System prompts the user to send another invite or invite another artist	

8. REVIEW A TRACK

USE CASE	8	Use Case 8 Review a Track
Description of Goal in Context	An artist opens a track shared by a music producer, listens to it in the virtual 3D studio with spatial audio, and leaves feedback, anywhere at any time.	
Preconditions	• The system app needs to be installed; • The artist accepted an invitation to collaborate; • The shared track is stored in the Tracks Database; • An internet connection is available.	
Post Conditions, Success End Condition	• The artist feedback is saved with the track; • The music producer is notified of the new feedback.	
DESCRIPTION of Scenario	• Artist opens the shared project, selecting the track to review; • System requests the track from the Tracks Database; • System displays the track as a floating row in the 3D studio; • Artist plays the track through spatial audio; • Artist leaves feedback, such as a comment or a rating; • System automatically saves the feedback, notifying the music producer.	

Main Flow

Step	**Action**	**Alternate**
8.1	Artist opens the app	
8.2	Artist opens the shared project, selecting the track to review	
8.3	System sends request to the Tracks Database	EF 8.3
8.4	Tracks Database returns the track	EF 8.4

8.5	System displays the track as a floating row in the 3D studio	
8.6	Artist presses play to listen to the track through spatial audio	AVF 8.6
8.7	Artist leaves feedback, such as a comment or a rating	
8.8	System saves feedback with the track, notifying music producer user	
8.9	System automatically saves the project	
EXCEPTIONS or ERROR Flow **Description:** Tracks Database unreachable or no internet connection		
Step	**Branching Action**	**Alternate**
8.3.1	System informs the user with an error message that the database is unreachable or there is no internet connection	
8.3.2	System offers to open locally stored projects in offline mode	
8.3.3	System automatically retries the connection	
EXCEPTIONS or ERROR Flow **Description:** Track no longer available or access removed		
Step	**Branching Action**	**Alternate**
8.4.1	System informs the artist user with an error message that the track is no longe available or the access was removed	
8.4.2	System prompts the artist user to contact the music producer user	
ALTERNATIVE or VARIATION Flow **Description:** User prefers touch or mouse controls		
Step	**Branching Action**	**Alternate**

8.6.1	In terms of accessibility, user prefers to use physical controls, instead of pinch and dragging gestures	
8.6.2	User types or speaks their choice to the system	
8.6.3	System listens and switches controls	

12.

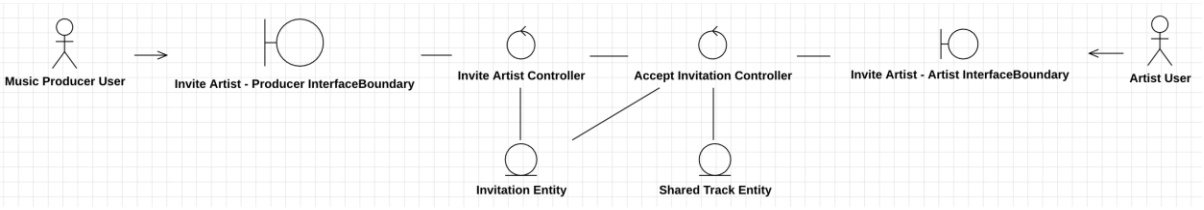
ANALYSIS CLASS 1

Analysis class diagrams represent a high conceptual level object to investigate, interpret, and break down complex systems to understand what it really needs from a different perspective.

These Analysis Classes were identified from the previous Use Case Diagrams, and will be used to identify Design Classes. Based on the three main types, entity, control, and boundary, they were developed on StarUML, focusing on the relationships between actors and databases.

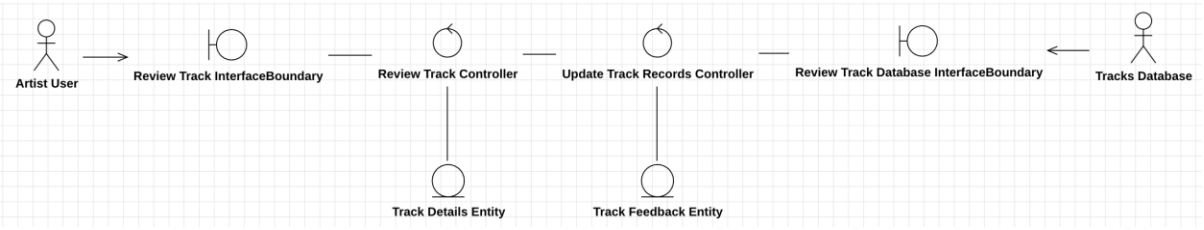
The entity class represents conceptual data model classes, the control class represents behaviours of specific use cases, and boundary class represents connections between actors and these use cases.

- **Chosen Use Case 1: Music Producer User > Use Case 7 (Invite Artist to Collaborate on Track) > Artist User**



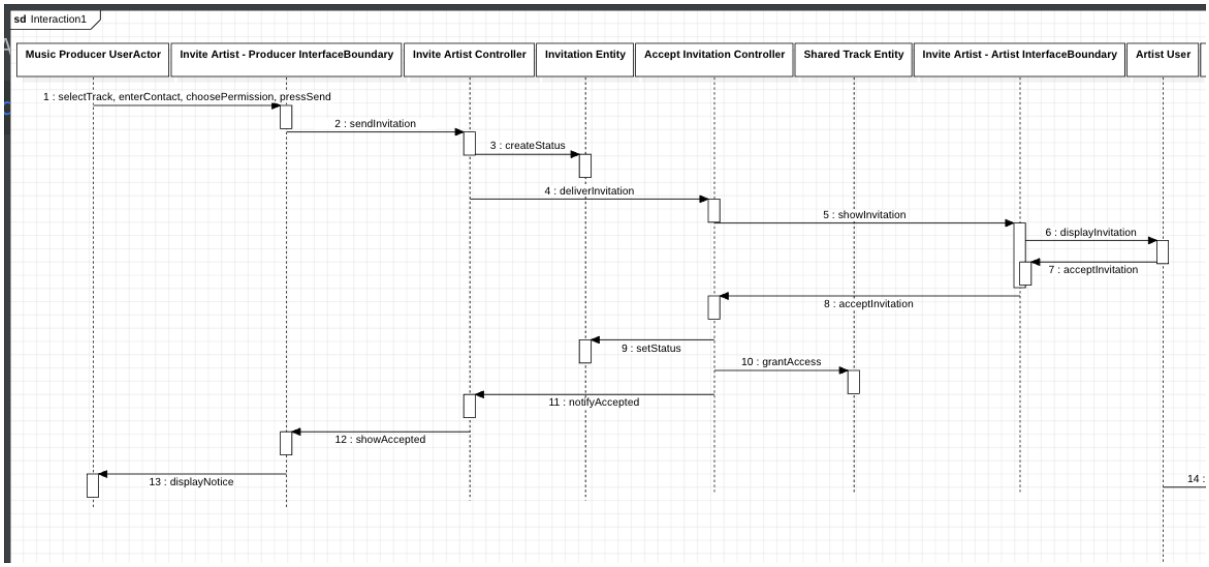
ANALYSIS CLASS 2

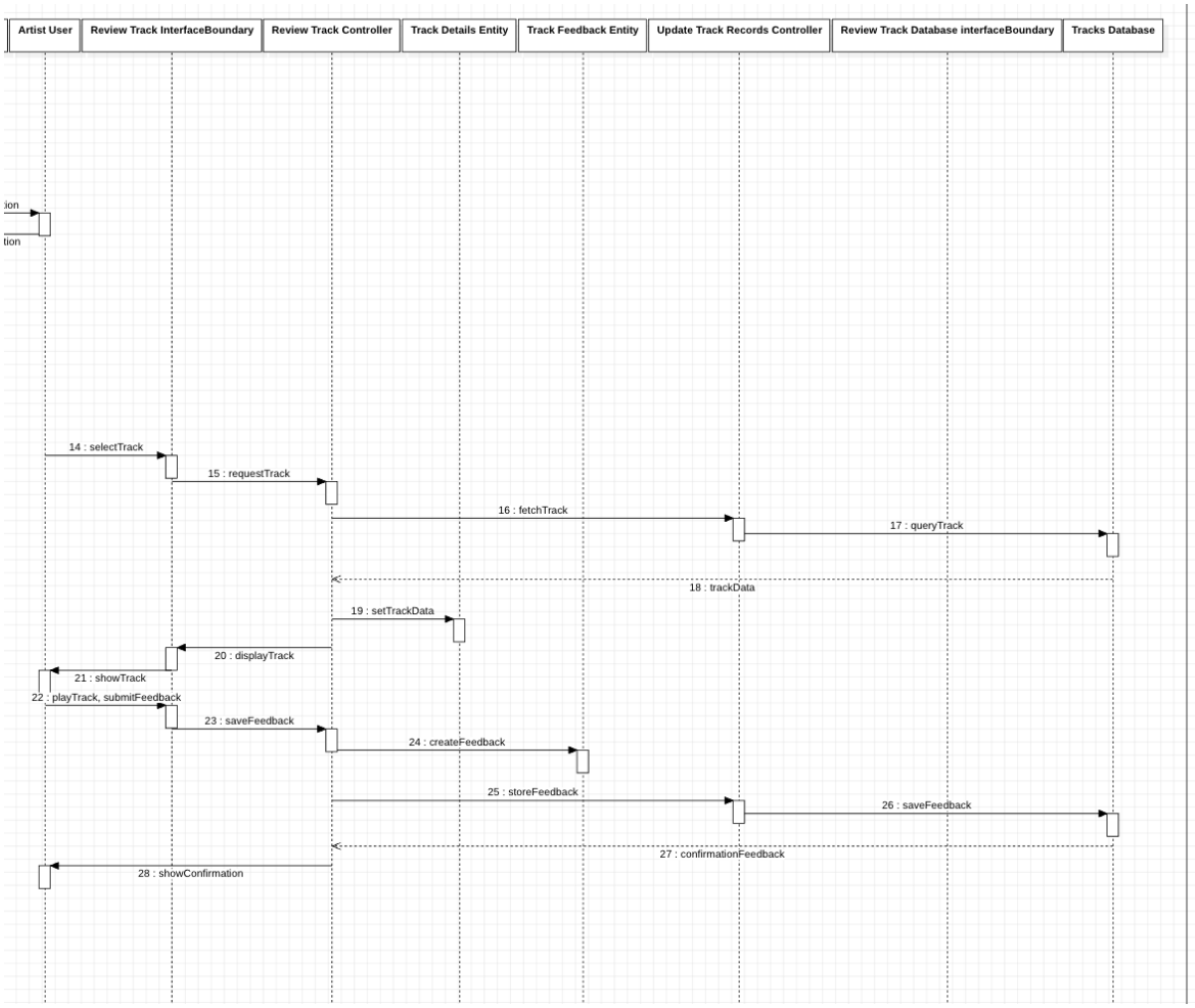
- **Chosen Use Case 2: Artist User > Use Case 8 (Review a Track) > Tracks Database**



SEQUENCE DIAGRAM 1

This is a communication and collaboration diagram, in the form of a sequence diagram, to analyse the main flows for the two previous analysis classes use cases under consideration





By identifying the design classes from the analysis classes, the following class diagram si being developed:

